

Jacob Keller

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Education

Olin College of Engineering – Needham, MA

May 2028

- BS, Electrical and Computer Engineering
- GPA: 4.0
- Relevant Coursework: Software Systems (**C**), Software Design (**Python**), Electronics, Modeling and Simulation (**MATLAB**), Principles of Integrated Engineering, Linear Algebra, Multivariable Calculus, Differential Equations

Experience

Low Voltage Lead, Olin Formula SAE Electric – Needham, MA

May 2025 – Present

- Led 5 engineers across 9 projects spanning **C** firmware for STM32, PCB design (**KiCAD**), and **Python** telemetry
- Re-factored legacy CAN library (**C**, **Bazel**) to transition from 8-bit AVR to 32-bit ARM architecture (STM32)
- Unified throttle and brake PCBs into a single custom VCU PCB, designed in **KiCAD**, reducing space claim by 40%

EV Engineering Intern, Formosa EV – Orange, CA

Mar 2025 – Aug 2025

- First intern and early employee at startup developing modern EV conversion components
- Flashed and tested firmware updates on MCU/SoC processors, documenting errors to support development team
- Designed high-voltage bus bars and sheet metal brackets in **Onshape** for motor, battery, and radiator integration.
- Adapted original throttle linkage from 1965 Corvair to modern electronic sensor, reducing cost with **3D-printed** parts

Integration Engineer, Olin Formula SAE Electric – Needham, MA

Sept 2024 – May 2025

- Developed embedded **C** firmware for ATmega16M1 microcontroller. Utilized CAN, SPI, and PWM protocols to measure and visually communicate low voltage system status and control the pump/fan within the cooling loop
- Created **Python** telemetry system using UART radios, enabling wireless diagnostics for the first time in team history

Information Technology Helpdesk Staff, Olin College – Needham, MA

Sept 2024 – Present

- Delivered technical support to a diverse user base of 450+ students, faculty, and staff as a Dell Certified Technician
- Diagnosed and resolved hardware/software issues across multiple operating systems and laptop configurations

Projects

Pedestrian Detection Algorithm, jacobkeller.me/ped

Dec 2024

- Built a **MATLAB** algorithm to detect the presence of a person in an image using principal component analysis (PCA)
- Achieved 78% classification accuracy, demonstrating PCA's effectiveness in improving computational efficiency while maintaining reasonable accuracy for pedestrian detection
- Presented findings to Olin College faculty and students through a professional academic poster

Carbon Dioxide Measurement Station, jacobkeller.me/co2

Aug 2024

- Designed a low-cost solar-powered device for measuring carbon cycling in scientific research
- Prototyped and refined electronic design through breadboard testing and final soldered assembly
- Developed **Python** software for an ESP32 microcontroller using PWM, SPI, I2C, and UART protocols to interface with environmental sensors, a real-time clock, and a linear actuator, while storing collected data on a microSD card

Skills and Interests

Programming: C, C++, Python, MATLAB, Bash

Software: Git, Linux, ROS2, Bazel, CMake, KiCAD, OnShape, Solidworks

Manufacturing: 3D Printing, Welding (TIG, Handheld Laser)

Interests: Backpacking, Rock Climbing, EV Conversions, Open-Source Hardware, Walkable Cities, History